

$$\begin{aligned}
& \frac{1}{16\pi^2} \left(\frac{1}{9} g_2^4 LF_{3,0}[m_2] \delta_{i1i2} + \frac{1}{6} g_2^4 LF_{4,-1}[m_2] \delta_{i1i2} - \frac{8}{45} g_2^4 LF_{5,-2}[m_2] \delta_{i1i2} + \frac{1}{18} \sum_p g_2^4 LF_{3,0}[m_q^p] \right. \\
& \delta_{i1i2} - \frac{5}{48} \sum_p g_2^4 LF_{4,-1}[m_q^p] \delta_{i1i2} + \frac{2}{45} \sum_p g_2^4 LF_{5,-2}[m_q^p] \delta_{i1i2} + \frac{1}{6} \sum_p g_2^4 LF_{3,0}[m_q^p] \delta_{i1i2} - \\
& \frac{5}{16} \sum_p g_2^4 LF_{4,-1}[m_q^p] \delta_{i1i2} + \frac{2}{15} \sum_p g_2^4 LF_{5,-2}[m_q^p] \delta_{i1i2} + \frac{1}{18} g_2^4 LF_{3,0}[\tilde{\mu}] \delta_{i1i2} + \\
& \frac{1}{12} g_2^4 LF_{4,-1}[\tilde{\mu}] \delta_{i1i2} - \frac{4}{45} g_2^4 LF_{5,-2}[\tilde{\mu}] \delta_{i1i2} + \frac{1}{18} g_1^2 \overline{y_d}^{i2p} y_d^{i1p} LF_{2,1,0}[m_1, m_d^p] - \\
& \frac{1}{9} g_1^2 \overline{y_d}^{i2p} y_d^{i1p} LF_{3,1,-1}[m_1, m_d^p] + \frac{1}{18} g_1^2 \overline{y_d}^{i2p} y_d^{i1p} LF_{4,1,-2}[m_1, m_d^p] + \\
& \frac{1}{864} g_1^2 g_2^2 LF_{2,1,0}[m_1, m_q^{i1}] \delta_{i1i2} + \frac{1}{864} g_1^2 g_2^2 LF_{2,2,-1}[m_1, m_q^{i1}] \delta_{i1i2} - \\
& \frac{1}{432} g_1^2 g_2^2 LF_{3,1,-1}[m_1, m_q^{i1}] \delta_{i1i2} + \frac{1}{864} g_1^2 g_2^2 LF_{4,1,-2}[m_1, m_q^{i1}] \delta_{i1i2} + \\
& \frac{1}{864} g_1^2 g_2^2 LF_{2,1,0}[m_1, m_q^{i2}] \delta_{i1i2} + \frac{1}{864} g_1^2 g_2^2 LF_{2,2,-1}[m_1, m_q^{i2}] \delta_{i1i2} - \\
& \frac{1}{432} g_1^2 g_2^2 LF_{3,1,-1}[m_1, m_q^{i2}] \delta_{i1i2} + \frac{1}{864} g_1^2 g_2^2 LF_{4,1,-2}[m_1, m_q^{i2}] \delta_{i1i2} - \\
& \frac{1}{96} g_2^4 LF_{2,1,0}[m_2, m_q^{i1}] \delta_{i1i2} - \frac{1}{96} g_2^4 LF_{2,2,-1}[m_2, m_q^{i1}] \delta_{i1i2} - \frac{5}{48} g_2^4 LF_{3,1,-1}[m_2, m_q^{i1}] \delta_{i1i2} + \\
& \frac{1}{32} g_2^4 LF_{4,1,-2}[m_2, m_q^{i1}] \delta_{i1i2} - \frac{1}{96} g_2^4 LF_{2,1,0}[m_2, m_q^{i2}] \delta_{i1i2} - \frac{1}{96} g_2^4 LF_{2,2,-1}[m_2, m_q^{i2}] \delta_{i1i2} - \\
& \frac{5}{48} g_2^4 LF_{3,1,-1}[m_2, m_q^{i2}] \delta_{i1i2} + \frac{1}{32} g_2^4 LF_{4,1,-2}[m_2, m_q^{i2}] \delta_{i1i2} - \frac{1}{3} g_2^4 LF_{2,1,0}[m_2, \tilde{\mu}] \delta_{i1i2} - \\
& \frac{1}{3} g_2^4 LF_{2,2,-1}[m_2, \tilde{\mu}] \delta_{i1i2} + \frac{1}{6} g_2^4 LF_{3,1,-1}[m_2, \tilde{\mu}] \delta_{i1i2} + \frac{1}{6} g_2^4 LF_{3,2,-2}[m_2, \tilde{\mu}] \delta_{i1i2} + \\
& \frac{2}{3} g_3^2 \overline{y_d}^{i2p} y_d^{i1p} LF_{2,1,0}[m_3, m_d^p] - \frac{4}{3} g_3^2 \overline{y_d}^{i2p} y_d^{i1p} LF_{3,1,-1}[m_3, m_d^p] + \\
& \frac{2}{3} g_3^2 \overline{y_d}^{i2p} y_d^{i1p} LF_{4,1,-2}[m_3, m_d^p] + \frac{1}{18} g_2^2 g_3^2 LF_{2,1,0}[m_3, m_q^{i1}] \delta_{i1i2} + \\
& \frac{1}{18} g_2^2 g_3^2 LF_{2,2,-1}[m_3, m_q^{i1}] \delta_{i1i2} - \frac{1}{9} g_2^2 g_3^2 LF_{3,1,-1}[m_3, m_q^{i1}] \delta_{i1i2} + \\
& \frac{1}{18} g_2^2 g_3^2 LF_{4,1,-2}[m_3, m_q^{i1}] \delta_{i1i2} + \frac{1}{18} g_2^2 g_3^2 LF_{2,1,0}[m_3, m_q^{i2}] \delta_{i1i2} + \\
& \frac{1}{18} g_2^2 g_3^2 LF_{2,2,-1}[m_3, m_q^{i2}] \delta_{i1i2} - \frac{1}{9} g_2^2 g_3^2 LF_{3,1,-1}[m_3, m_q^{i2}] \delta_{i1i2} + \\
& \frac{1}{18} g_2^2 g_3^2 LF_{4,1,-2}[m_3, m_q^{i2}] \delta_{i1i2} + \frac{1}{8} g_2^2 \overline{a_e}^{pr} a_e^{pr} LF_{4,1,-1}[m_q^p, m_e^r] \delta_{i1i2} - \\
& \frac{1}{6} g_2^2 \overline{a_e}^{pr} a_e^{pr} LF_{5,1,-2}[m_q^p, m_e^r] \delta_{i1i2} + \frac{3}{8} g_2^2 \overline{a_d}^{pr} a_d^{pr} LF_{4,1,-1}[m_q^p, m_d^r] \delta_{i1i2} - \\
& \frac{1}{2} g_2^2 \overline{a_d}^{pr} a_d^{pr} LF_{5,1,-2}[m_q^p, m_d^r] \delta_{i1i2} + \frac{1}{4} g_2^2 \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{pr} LF_{2,2,0}[m_q^p, m_u^r] \delta_{i1i2} - \\
& \frac{1}{8} g_2^2 \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{pr} LF_{3,2,-1}[m_q^p, m_u^r] \delta_{i1i2} - \frac{1}{432} g_1^2 g_2^2 LF_{2,1,0}[m_q^{i1}, m_1] \delta_{i1i2} + \\
& \frac{1}{864} g_1^2 g_2^2 LF_{3,1,-1}[m_q^{i1}, m_1] \delta_{i1i2} + \frac{1}{48} g_2^4 LF_{2,1,0}[m_q^{i1}, m_2] \delta_{i1i2} - \\
& \frac{1}{96} g_2^4 LF_{3,1,-1}[m_q^{i1}, m_2] \delta_{i1i2} - \frac{1}{9} g_2^2 g_3^2 LF_{2,1,0}[m_q^{i1}, m_3] \delta_{i1i2} + \\
& \frac{1}{18} g_2^2 g_3^2 LF_{3,1,-1}[m_q^{i1}, m_3] \delta_{i1i2} - \frac{1}{432} g_1^2 g_2^2 LF_{2,1,0}[m_q^{i2}, m_1] \delta_{i1i2} + \\
& \frac{1}{864} g_1^2 g_2^2 LF_{3,1,-1}[m_q^{i2}, m_1] \delta_{i1i2} + \frac{1}{48} g_2^4 LF_{2,1,0}[m_q^{i2}, m_2] \delta_{i1i2} - \\
& \frac{1}{96} g_2^4 LF_{3,1,-1}[m_q^{i2}, m_2] \delta_{i1i2} - \frac{1}{9} g_2^2 g_3^2 LF_{2,1,0}[m_q^{i2}, m_3] \delta_{i1i2} + \\
& \frac{1}{18} g_2^2 g_3^2 LF_{3,1,-1}[m_q^{i2}, m_3] \delta_{i1i2} - \frac{1}{4} g_2^2 \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{pr} LF_{3,1,0}[m_u^r, m_q^p] \delta_{i1i2} - \\
& \frac{1}{4} g_2^2 \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{pr} LF_{3,2,-1}[m_u^r, m_q^p] \delta_{i1i2} + \frac{3}{4} g_2^2 \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{pr} LF_{4,1,-1}[m_u^r, m_q^p] \delta_{i1i2} - \\
& \frac{1}{2} g_2^2 \tilde{\mu}^2 \overline{y_u}^{pr} y_u^{pr} LF_{5,1,-2}[m_u^r, m_q^p] \delta_{i1i2} - \frac{5}{24} g_1^2 g_2^2 LF_{4,1,-2}[\tilde{\mu}, m_1] \delta_{i1i2} + \\
& \frac{1}{6} g_1^2 g_2^2 LF_{5,1,-3}[\tilde{\mu}, m_1] \delta_{i1i2} + \frac{2}{3} g_2^4 LF_{3,1,-1}[\tilde{\mu}, m_2] \delta_{i1i2} + \\
& \frac{1}{3} g_2^4 LF_{3,2,-2}[\tilde{\mu}, m_2] \delta_{i1i2} - \frac{9}{8} g_2^4 LF_{4,1,-2}[\tilde{\mu}, m_2] \delta_{i1i2} + \frac{1}{2} g_2^4 LF_{5,1,-3}[\tilde{\mu}, m_2] \delta_{i1i2} - \\
& \frac{1}{8} g_2^2 \overline{y_d}^{i2p} y_d^{i1p} LF_{3,1,-1}[\tilde{\mu}, m_d^p] + \frac{1}{24} g_2^2 \overline{y_d}^{i2p} y_d^{i1p} LF_{4,1,-2}[\tilde{\mu}, m_d^p] + \\
& \frac{1}{2} \overline{y_d}^{rp} \overline{y_d}^{i2s} y_d^{rs} y_d^{i1p} LF_{2,1,0}[\tilde{\mu}, m_q^r] - \overline{y_d}^{rp} \overline{y_d}^{i2s} y_d^{rs} y_d^{i1p} LF_{3,1,-1}[\tilde{\mu}, m_q^r] + \\
& \frac{1}{2} \overline{y_d}^{rp} \overline{y_d}^{i2s} y_d^{rs} y_d^{i1p} LF_{4,1,-2}[\tilde{\mu}, m_q^r] - \frac{1}{8} g_2^2 \overline{y_u}^{i2p} y_u^{i1p} LF_{3,1,-1}[\tilde{\mu}, m_u^p] + \\
& \frac{1}{24} g_2^2 \overline{y_u}^{i2p} y_u^{i1p} LF_{4,1,-2}[\tilde{\mu}, m_u^p] + \frac{1}{72} m_1 g_1^2 \overline{y_d}^{i2p} a_d^{i1p} LF_{2,1,1,0}[m_1, m_d^p, m_q^{i1}] + \\
& \frac{1}{144} m_1 g_1^2 \overline{y_d}^{i2p} a_d^{i1p} LF_{2,2,1,-1}[m_1, m_d^p, m_q^{i1}] - \frac{1}{72} m_1 g_1^2 \overline{y_d}^{i2p} a_d^{i1p} LF_{3,1,1,-1}[m_1, m_d^p, m_q^{i1}] + \\
& \frac{1}{72} m_1 g_1^2 y_d^{i1p} \overline{a_d}^{i2p} LF_{2,1,1,0}[m_1, m_d^p, m_q^{i2}] + \frac{1}{144} m_1 g_1^2 y_d^{i1p}$$